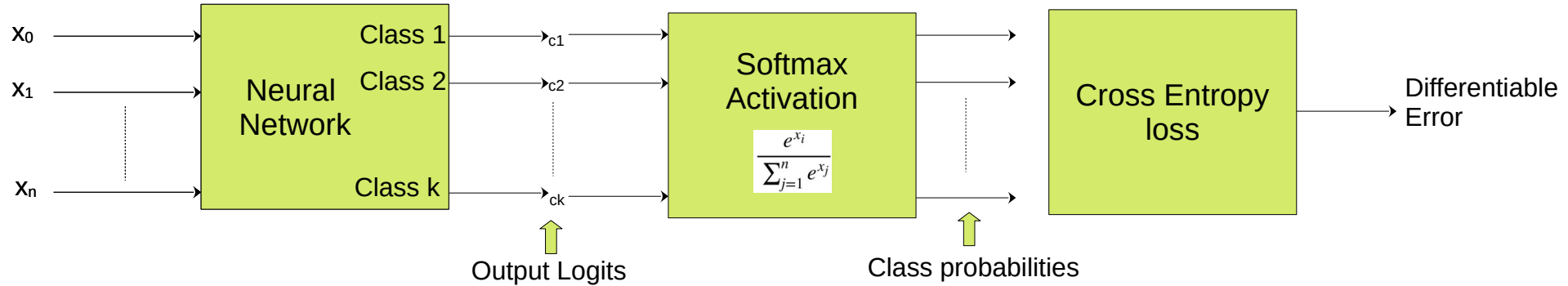


Softmax and Cross Entropy in Classification

Goals:

- Where they fit in a classifier pipeline
- Explain what softmax does and why we use it
- Explain cross-entropy and why we use it
- Implement the pair correctly for classification

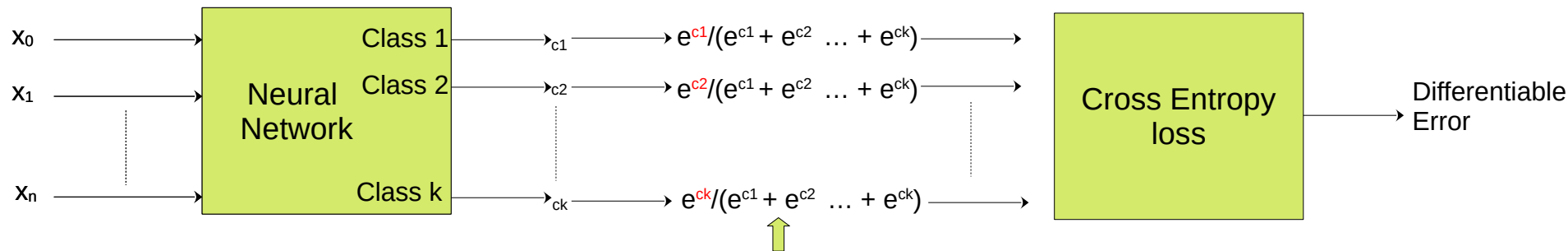
Where They Fit in a Classifier



- **Logits**: predicted class scores from model (unnormalized)
- Largest logit is the winner
- **Softmax** outputs class 'probabilities'
- **Cross Entropy** how wrong we are.

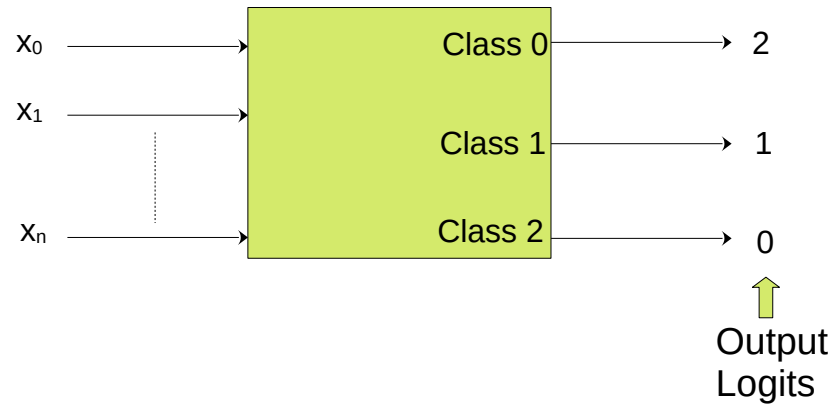
Note: Softmax and cross entropy are used ONLY for training.
For evaluation (inferencing) use argmax on the output logits

Softmax Activation Explanation

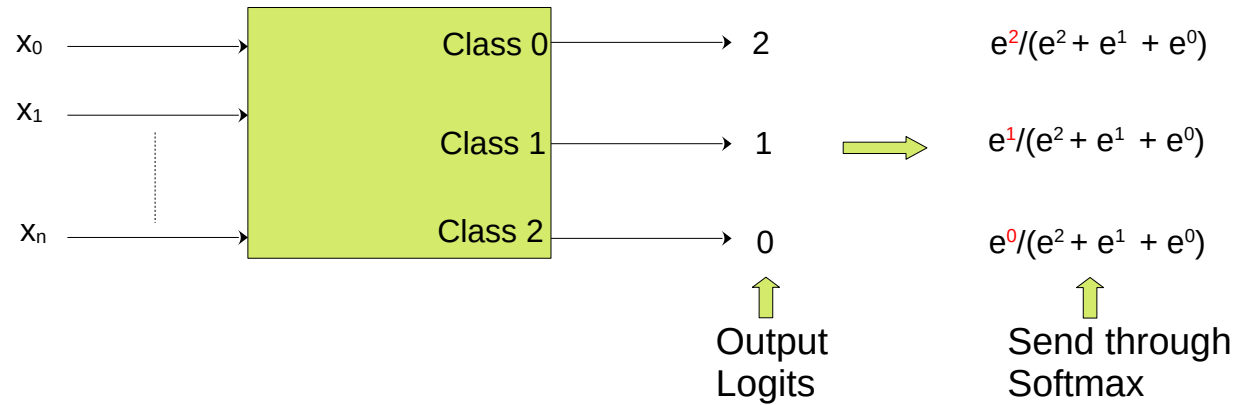


- Turns raw scores (“logits”) into a ‘probability’ like distribution (sum to 1)
- **Differentiable**
- Largest softmaxed logit is still largest
- In fact softmax pushes largest higher and others lower

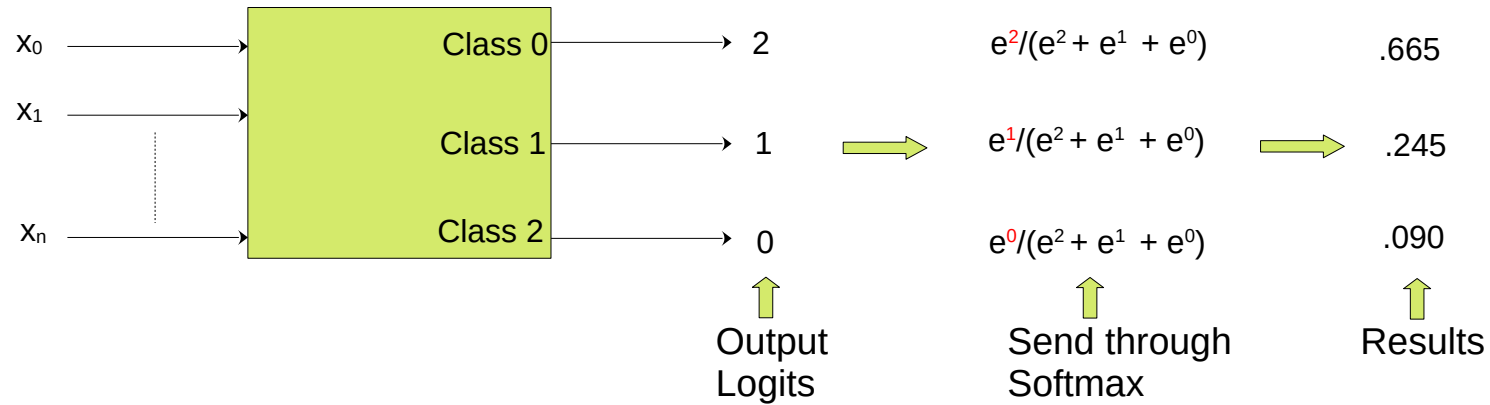
Softmax Activation Example



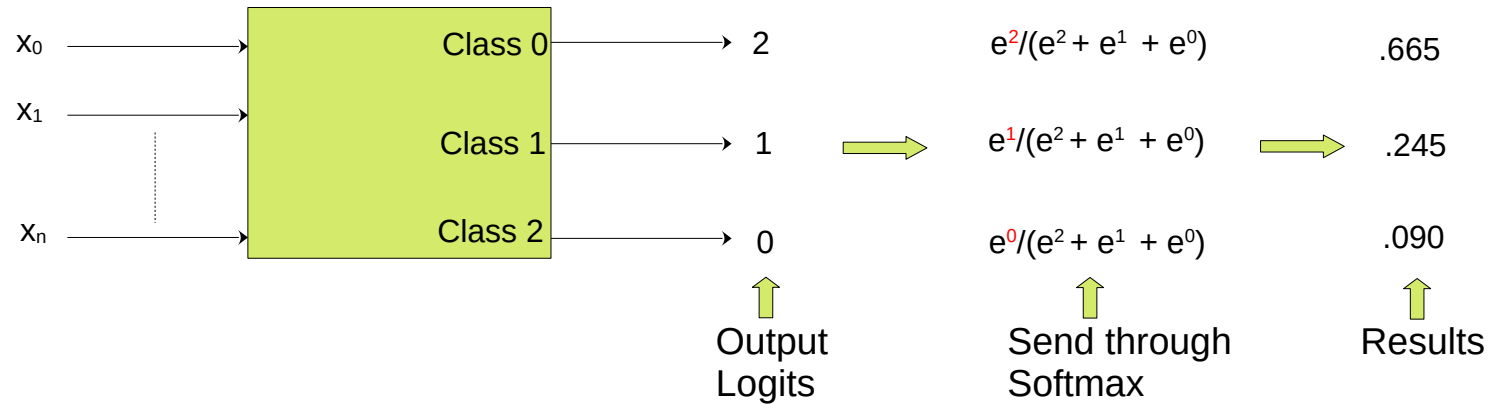
Softmax Activation Example



Softmax Activation Example



Softmax Activation Example



- $.665 + .245 + .090 = 1$
- Class 0 is still selected
- Class 0 probability is much larger than the original gap between Class 0 and Class 1 suggests

Classification Example

Use argmax to select predicted class (bad idea)



A NN with a final layer softmax activation function.
Three training rows are fed in with the following results

c1	c2	c3	targets	correct?
0.3	0.3	0.4	0 0 1 (democrat)	yes
0.3	0.4	0.3	0 1 0 (republican)	yes
0.1	0.2	0.7	1 0 0 (other)	no

It has a classification accuracy of 2/3 or 66%

It has a classification error of 1/3 or 33%

It barely gets the first 2 right and is way off on the third.

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Use argmax to select predicted class (bad idea)



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It barely gets the first 2 right and is way off on the third.

A different NN with a final layer softmax activation function.
Three training rows are fed in with the following results

c1	c2	c3	targets	correct?
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It nails the first 2 and barely misses the third.

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The second network is clearly better but the metrics do not reflect that
Also, argmax is not differentiable, cannot backprop through it

Cannot use argmax

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Classification Example

Use Cross Entropy Classification Error

- Also called the **negative log** likelihood
- Takes the negative log of the softmax score for ONLY the correct class. All other entries are ignored.
- So applying cross entropy error for the previous 2 examples;

c1	c2	c3	targets	correct?
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0.3	0.4	0.3	0 1 0 (republican)	yes
0.1	0.2	0.7	1 0 0 (other)	no

Row1 error= $-(\ln(0.3)*0) + (\ln(0.3)*0) + (\ln(0.4)*1) = -\ln(0.4)$

Row2 error= $-\ln(.4)$

Row3 error= $-\ln(.1)$

Total error= $-(\ln(0.4) + \ln(0.4) + \ln(0.1)) / 3 = 1.38$

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Row1 error= $-(\ln(0.1)*0) + (\ln(0.2)*0) + (\ln(0.7)*1) = -\ln(0.7)$

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Row3 error= $-\ln(.3)$

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Total error= $-(\ln(0.7) + \ln(0.7) + \ln(0.3)) / 3 = 0.64$



The error now reflects that the second network is much better than the first

Pytorch Cross Entropy Example

```
criterion = nn.CrossEntropyLoss()
```

```
def train(model, train_loader, test_loader, optimizer, epochs=10):
    for ep in range(1, epochs+1):
        model.train()
        run_loss, n = 0.0, 0
        for xb, yb in train_loader:
            xb, yb = xb.to(device), yb.to(device)
            optimizer.zero_grad(set_to_none=True)
            logits = model(xb)
            loss = criterion(logits, yb)
            loss.backward()
```

Note: pytorch cross entropy will automatically apply softmax before it calculates cross entropy loss (SO DONT ADD A SOFTMAX LAYER)

When not to use Cross Entropy

- Multi-label problems (you can have more than one label)
- Instead feed logits into BCEWithLogitsLoss- It will first apply a sigmoid activation to each output)

Summary

When Training:

- Softmax and why (converts to probabilities, differentiable)
- For single label classification use cross entropy loss (pytorch implementation automatically applies softmax to the input logits)
- For multilabel classification use BCEWithLogitsLoss

When Testing:

- **use argmax on the output logits**